

Exercise J01

Q1. The area between the curve $xy = a^2$, x-axis, $x = a$ and $x = 2a$ is-

- (A) $a \log 2$ (B) $a^2 \log 2$ (C) $2a \log 2$ (D) None of these

Ans. B

Sol. -

Q2. Area under the curve $y = \sin 2x + \cos 2x$ between $x = 0$ and $x = \frac{\pi}{4}$, is-

- (A) 2 sq. units (B) 1 sq. units (C) 3 sq. units (D) 4 sq. units

Ans. B

Sol. -

Q3. The area bounded between the curve $|y| = 1 - x^2$ is-

- (A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) None of these

Ans. C

Sol. -

Q4. The area bounded by the parabola $y^2 = 4ax$, x-axis and $x = 2a$ and latus rectum is-

- (A) $2a^2 (\sqrt{2} - 1)$ (B) $4a^2 (2\sqrt{2} - 1)$ (C) $\frac{4}{3}a^2 (2\sqrt{2} - 1)$ (D) $\frac{8a^2}{3} (2\sqrt{2} - 1)$

Ans. C

Sol. -

Q5. The area bounded between parabola $x^2 = 4y$ and $y = |x|$ is -

- (A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) $\frac{16}{3}$

Ans. D

Sol. -

Q6. The area of the figure bounded by the parabola $y = x^2 + 1$ and the straight line $x + y = 3$ is-

- (A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{7}{2}$ (D) $\frac{9}{2}$

Ans. D

Sol. -

Q7. Common area between the parabolas $y = 2x^2$ and $y = x^2 + 4$ is-

- (A) $\frac{16}{3}$ (B) $\frac{8}{3}$ (C) $\frac{32}{3}$ (D) None of these

Ans. C

Sol. -

Q8. Area bounded by the curves $y = 2^x$, $y = 2x - x^2$, $x = 0$ and $x = 2$ is-

- (A) $\frac{3}{\log 2} - \frac{4}{3}$ (B) $\frac{3}{\log 2} + \frac{4}{3}$ (C) $3 \log 2 + \frac{4}{3}$ (D) $3 \log 2 - \frac{4}{3}$

Ans. A

Sol. -

Q9. The area of the closed figure bounded by $y = \cos x$, $y = 1 + (2/\pi)x$ and $x = \pi/2$ is-

- (A) $3\pi/4$ (B) $\pi/2$ (C) $\frac{3\pi - 4}{4}$ (D) $\frac{\pi - 4}{4}$

Ans. C

Sol. -

Q10. The area enclosed between the curves $y = \log_e(x + e)$, $x = \log_e\left(\frac{1}{y}\right)$ and the x-axis is

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. -

Q11. Area of the circle $(x - 2)^2 + (y - 3)^2 = 32$ which lies below the line $y = x + 1$ is-

- (A) $\int_{-2}^6 \left[(x + 1) + \sqrt{32 - (x - 2)^2 + 3} \right] dx$ (B) $\int_{-2}^6 \left[\left\{ 32 - (x - 2)^2 + 3 \right\} - (x + 1) \right] dx$

- (C) 16π (D) None of these

Ans. C

Sol. -

Q12. If $y = 2 \sin x + \sin 2x$ for $0 \leq x \leq 2\pi$, then the area enclosed by the curve and the x-axis is:

- (A) 6 (B) 8 (C) 12 (D) None

Ans. B

Sol. -

Q13. The area of the closed figure bounded by $x = -1$, $x = 2$ & $y = \begin{cases} -x^2 + 2 & , x \leq 1 \\ 2x - 1 & , x > 1 \end{cases}$ & the abscissa axis, is

- (A) $\frac{7}{3}$ (B) $\frac{10}{3}$ (C) $\frac{13}{3}$ (D) $\frac{16}{3}$

Ans. D

Sol. -

Q14. Area between curve $y = \frac{x}{1+x^2}$ and x axis from $x = -1$ to $x = 1$ is

- (A) $\ln 2$ (B) $\ln 4$ (C) $\pi/2$ (D) π

Ans. A

Sol. -

Q15. The area of the figure bounded by the curves $y = e^x$, $y = e^{-x}$ & the straight line $x = 1$ is

- (A) $e + \frac{1}{e}$ (B) $e - \frac{1}{e}$ (C) $e + \frac{1}{e} - 2$ (D) None

Ans. C

Sol. -

Q16. The area bounded by $y = x^2$ & $y = 2 - x^2$ is:

- (A) $\frac{8}{3}$ (B) $\frac{16}{3}$ (C) $\frac{32}{3}$ (D) $\frac{64}{3}$

Ans. A

Sol. -

Q17. The area bounded by the curves $y^2 = 3x$ & $y^2 = 36 - 9x$ is:

- (A) 16 (B) 18 (C) 21.5 (D) none

Ans. A

Sol. -

Q18. The ratio in which the area enclosed by the curve $y = \cos x$ $\left(0 \leq x \leq \frac{\pi}{2}\right)$ in the first quadrant is divided by the curve $y = \sin x$, is:

- (A) $(\sqrt{2} - 1) : 1$ (B) $(\sqrt{2} + 1) : 1$ (C) $\sqrt{2} : 1$ (D) None

Ans. C

Sol. -

Q19. The area bounded by the curve $y^2 + x^4 = x^2$ is:

- (A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) None

Ans. B

Sol. -

Q20. Area bounded by the curves $y = |x| - 2$ and $y = 1 - |x-1|$ is equal to

- (A) 4 sq. units (B) 6 sq. units (C) 2 sq. units (D) 8 sq. units

Ans. A

Sol. -

Q21. Area bounded by the parabola $y = x^2 - 2x + 3$ and tangents drawn to it from the point $P(1, 0)$ is equal to

- (A) $4\sqrt{2}$ sq. units (B) $\frac{4\sqrt{2}}{3}$ sq. units (C) $\frac{8\sqrt{2}}{3}$ sq. units (D) $\frac{16}{3}\sqrt{2}$ sq. units

Ans. B

Sol. -

Q22. Area bounded by the curves $y = \sin x$, tangent drawn to it at $x = 0$ and the line $x = \frac{\pi}{2}$, is equal to

- (A) $\frac{\pi^2 - 4}{2}$ sq. units (B) $\frac{\pi^2 - 8}{8}$ sq. units (C) $\frac{\pi^2 - 2}{4}$ sq. units (D) $\frac{\pi^2 - 2}{2}$ sq. units

Ans. B

Sol. -

Q23. Value of the parameter a such that the area bounded by $y = a^2 x^2 + ax + 1$, coordinate axes and the line $x = 1$, attains it's least value, is equal to

- (A) $-\frac{1}{4}$ (B) $-\frac{1}{2}$ (C) $-\frac{3}{4}$ (D) -1

Ans. C

Sol. -

Q24. Area enclosed by the curve $y = f(x)$ defined parametrically as $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$ is equal to

- (A) π sq. units (B) $\pi/2$ sq. units (C) $\frac{3\pi}{4}$ sq. units (D) $\frac{3\pi}{2}$ sq. units

Ans. A

Sol. -

Q25. The area of the region bounded by the curves $y = |x - 1|$ and $y = 3 - |x|$ is-

- (A) 6 sq. units (B) 2 sq. units (C) 3 sq. units (D) 4 sq. units

Ans. D

Sol. -

Q26. The area enclosed between the curve $y = \log_e (x + e)$ and the coordinate axes is -

(A) 1 (B) 2 (C) 3 (D) 4

Ans. A

Sol. -

Q27. The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the coordinate axes. If S_1 , S_2 , S_3 are respectively the areas of these parts numbered from top to bottom; then $S_1 : S_2 : S_3$ is -

(A) 1 : 2 : 1 (B) 1 : 2 : 3 (C) 2 : 1 : 2 (D) 1 : 1 : 1

Ans. D

Sol. -

Q28. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is

(A) 1 (B) 2 (C) $2\sqrt{2}$ (D) 4

Ans. B

Sol. -

Q29. If the area bounded by $y = ax^2$ and $x = ay^2$, $a > 0$, is 1, then $a =$

(A) 1 (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{\sqrt{3}}$

Ans. B

Sol. -

Q30. The area bounded by the parabolas $y = (x + 1)^2$ and $y = (x - 1)^2$ and the line $y = 1/4$ is

(A) 4 sq. units (B) 1/6 sq. units (C) 4/3 sq. units (D) 1/3 sq. units

Ans. D

Sol. -

Q31. The area of the region between the curves $y = \sqrt{\frac{1 + \sin x}{\cos x}}$ and $y = \sqrt{\frac{1 - \sin x}{\cos x}}$ bounded by the lines $x = 0$ and $x = \frac{\pi}{4}$ is

(A) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$ (B) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$ (C) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$
(D) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$

Ans. B

Sol.

$$\begin{aligned} \text{Area} &= \int_0^{X/4} \left(\sqrt{\frac{1+\sin x}{\cos x}} - \sqrt{\frac{1-\sin x}{\cos x}} \right) dx \\ &= \int_0^{X/4} \frac{\left(\cos \frac{x}{2} + \sin \frac{x}{2} \right) - \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)}{\sqrt{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}} dx \\ &= \int_0^{X/4} \frac{2\sin \frac{x}{2}}{\sqrt{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}} dx = \int_0^{X/4} \frac{2\tan \frac{x}{2}}{\sqrt{1 - \tan^2 \frac{x}{2}}} dx \end{aligned}$$

Let $\frac{x}{2} = t$

$$\sec^2 \frac{x}{2} dx = dt \Rightarrow dx = \frac{2dt}{(1+t^2)}$$

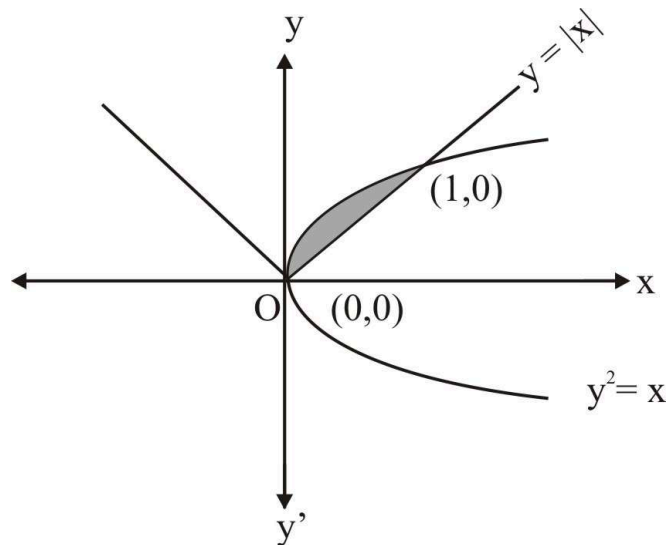
$$\therefore \text{Area} = \int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$$

Q32. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is

- (A) $\frac{2}{3}$ (B) 1 (C) $\frac{1}{6}$ (D) $\frac{1}{3}$

Ans. C

Sol.



$$\begin{aligned} \text{Required area} &= \int_0^1 (\sqrt{x} - x) dx = \left(\frac{2}{3}x^{3/2} - \frac{x^2}{2} \right)_0^1 \\ &= \frac{2}{3} - \frac{1}{2} = \frac{4-3}{6} = \frac{1}{6} \text{ sq unit} \end{aligned}$$

Q33. The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to -

- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{5}{3}$

Ans. C

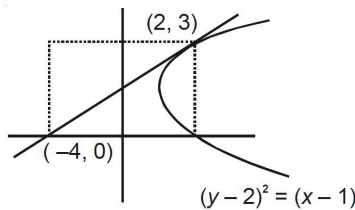
Sol. -

Q34. The area of the region bounded by the parabola $(y - 2)^2 = x - 1$, the tangent to the parabola at the point $(2, 3)$ and the x - axis is -

- (A) 3 (B) 6 (C) 9 (D) 12

Ans. C

Sol.



The equation of tangent at $(2, 3)$ to the given parabola is $x = 2y - 4$

Required area $\int_0^b \{(y - 2)2 + 1 - 2y + 4\} dy$

$$= \left[\frac{(y - 2)^3}{3} - y^2 + 5y \right]_0^3$$

$$= \frac{1}{3} - 9 + 15 + \frac{8}{5}$$

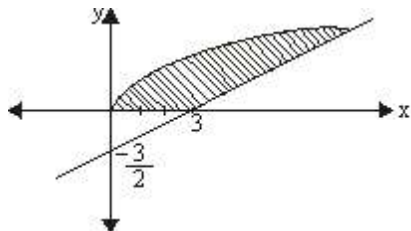
= 9 sq. units.

Q35. The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$ x -axis and lying in the first quadrant is :

- (A) 9 (B) 36 (C) 18 (D) $27/4$

Ans. A

Sol.



$$y = \sqrt{x} \quad \dots\dots\dots(1)$$

$$\text{and } 2y - x + 3 = 0 \quad \dots\dots\dots(2)$$

On solving both $y = -1, 3$

$$\text{Required area} = \int_0^2 \{(2y + 3) - y^2\} dy$$

$$\begin{aligned}
&= y^2 + 3y - \frac{y^3}{3} \Big|_0^3 \\
&= 9 + 9 - 9 \\
&= 9
\end{aligned}$$

Q36. Let $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, and α, β ($\alpha < \beta$) be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. unit) bounded by the curve $y = (g \circ f)(x)$ and the lines $x = \alpha$, $x = \beta$ and $y = 0$ is

- (A) $\frac{1}{2}(\sqrt{3} - \sqrt{2})$ (B) $\frac{1}{2}(\sqrt{2} - 1)$ (C) $\frac{1}{2}(\sqrt{3} - 1)$ (D) $\frac{1}{2}(\sqrt{3} + 1)$

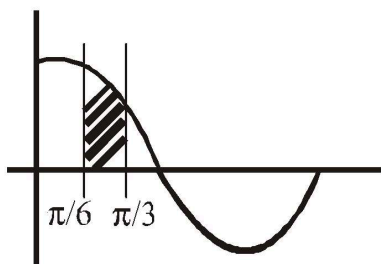
Ans. C

Sol. $g \circ f(x) = \cos x$

$$18x^2 - \pi x + \pi^2 = 0$$

$$(6x - \pi)(3x - \pi) = 0$$

$$x = \frac{\pi}{6}, x = \frac{\pi}{3}$$



$$A = \int_{\pi/6}^{\pi/3} \cos x \, dx = \frac{\sqrt{3} - 1}{2}$$

Paragraph

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of $f(x)$ and let $t = |s|$.

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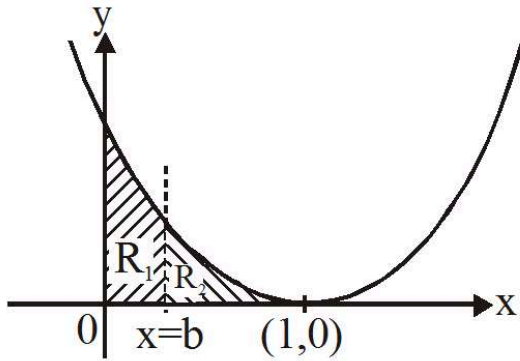
Q37. Let the straight line $x = b$ divide the area enclosed by $y = (1 - x^2)$, $y = 0$ and $x = 0$ into two parts R_1 ($0 \leq x \leq b$) and R_2 ($b \leq x \leq 1$) such that $R_1 - R_2 = \frac{1}{4}$. Then b equals

:

- (A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Ans. B

Sol.



$$\therefore R_1 - R_2 = \frac{1}{4}$$

$$\Rightarrow \int_0^b x^2(1-x)x^2 dx - \int_b^1 x^2(1-x)x^2 dx = \frac{1}{4}$$

$$\Rightarrow -\left(\frac{(1-x)^3}{3}\right)_0^b + \left(\frac{(1-x)^3}{3}\right)_0^b = \frac{1}{4}$$

$$\Rightarrow \left(\frac{(1-b)^3}{3} \cdot \frac{1}{3}\right) - \frac{(1-b)^3}{3} = \frac{1}{4}$$

$$\Rightarrow \frac{1}{3} - \frac{2}{3}(1-b)^3 = \frac{1}{4} \Rightarrow \frac{2}{3}(1-b)^3 = \frac{1}{12}$$

$$\Rightarrow (1-b)^3 = \frac{1}{8} \Rightarrow 1-b = \frac{1}{2}$$

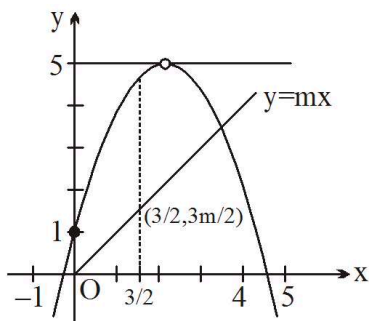
$$\Rightarrow b = \frac{1}{2}$$

Q38. The line $y = mx$ bisects the area enclosed by the curve $y = 1 + 4x - x^2$ & the lines $x = 0$, $x = 2$ & $y = 0$. Then the value of m is:

- (A) $\frac{13}{6}$ (B) $\frac{6}{13}$ (C) $\frac{11}{6}$ (D) none

Ans. C

Sol.



$$2. \frac{1}{2} \left(\frac{3}{2} \cdot \frac{3m}{2} \right) = \int_0^{3/2} (1 + 4x - x^2) dx$$

$$\frac{9m}{4} = x + 2x^2 - \frac{x^3}{3} \Big|_0^{3/2} = \frac{3}{2} + \frac{9}{2} - \frac{27}{8} \cdot \frac{1}{3} = 6 - \frac{9}{8} = \frac{39}{8}$$

$$m = \frac{13}{6} \text{ Ans.}$$

Q39. Area enclosed by the curve $|x - 2| + |y + 1| = 1$ is equal to

- (A) 4 sq. units (B) 6 sq. units (C) 2 sq. units (D) 8 sq. units

Ans. C

Sol. -

Q40. The area of the region of the plane bounded by $\max(|x|, |y|) \leq 1$ & $xy \leq \frac{1}{2}$ is:

- (A) $\frac{1}{2} + \ln 2$ (B) $\frac{31}{4}$ (C) $1 + 2 \ln 2$ (D) $3 + \ln 2$

Ans. D

Sol. -

Q41. Area of the region which consists of all the points satisfying the conditions $|x - y| + |x + y| \leq 8$ and $xy \leq 2$, is equal to

- (A) $4(7 - \ln 8)$ sq. units (B) $4(9 - \ln 8)$ sq. units (C) $2(7 - \ln 8)$ sq. units
(D) $2(9 - \ln 8)$ sq. units

Ans. A

Sol. -

Q42. A point P moves in xy plane in such a way that $[|x|] + [|y|] = 1$, where $[.]$ denotes the greatest integer function. Area of the region representing all possible positions of the point P is equal to

- (A) 4 sq. units (B) 16 sq. units (C) $2\sqrt{2}$ sq. units (D) 8 sq. units

Ans. D

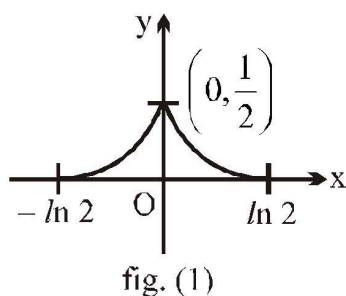
Sol. -

Q43. The area defined by $|y| \leq e^{-|x|} - \frac{1}{2}$ in cartesian co-ordinate system, is

- (A) $(4 - 2 \ln 2)$ (B) $(4 - \ln 2)$ (C) $(2 - \ln 2)$ (D) $(2 - 2 \ln 2)$

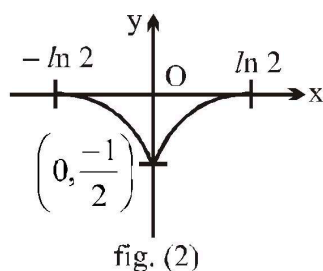
Ans. D

Sol.



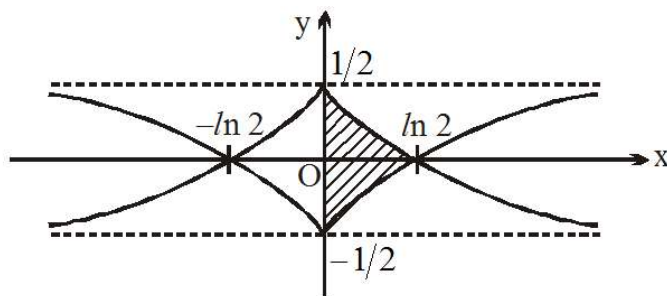
For $y \geq 0$

$$\therefore y = e^{-|x|} - \frac{1}{2} = \begin{cases} e^{-x} - \frac{1}{2} & \text{if } x \geq 0 \\ e^{-x} \frac{1}{2} & \text{if } x < 0 \end{cases} \quad \dots\dots\dots(1)$$



and for $y < 0$,

$$-y = \frac{1}{2} - e^{-|x|} = \begin{cases} \frac{1}{2} - e^{-x} & \text{if } x \geq 0 \\ \frac{1}{2} e^{-x} & \text{if } x < 0 \end{cases} \quad \dots\dots\dots(2)$$



Combining (1) and (2) we have the required are given as

$$A = 4 \int_0^{\ln 2} \left(e^{-x} - \frac{1}{2} \right) dx = 4 \left(\frac{1}{2} - \frac{1}{2} \ln 2 \right) = (2 - 2 \ln 2) \text{ Ans.}$$

Q44. The area of the region described by $A = \{(x, y) : x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1 - x\}$ is :

- (A) $\frac{\pi}{2} - \frac{2}{3}$ (B) $\frac{\pi}{2} + \frac{2}{3}$ (C) $\frac{\pi}{2} + \frac{4}{3}$ (D) $\frac{\pi}{2} - \frac{4}{3}$

Ans. C

Sol. Required area = $\pi - 2 \int_0^1 (\sqrt{1-x^2} - \sqrt{1-x}) dx$

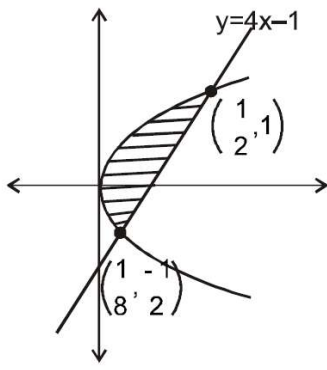
$$= \pi - 2 \left[\frac{x\sqrt{1-x^2}}{2} + \frac{1}{2} \sin^{-1} x + \frac{2}{3} (1-x)^{3/2} \right]_0^1 = \pi - 2 \left[\frac{\pi}{4} - \frac{2}{3} \right] = \frac{\pi}{2} + \frac{4}{3}$$

Q45. The area (in sq. units) of the region described by $[(x, y) : y^2 \leq 2x \text{ and } y \geq 4x - 1]$ is :

- (A) $\frac{1}{32}$ (B) $\frac{5}{64}$ (C) $\frac{15}{64}$ (D) $\frac{9}{32}$

Ans. D

Sol.



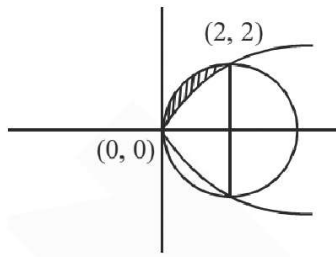
$$\begin{aligned}
 & \int_{-1/2}^2 \left(\frac{y+1}{4} - \frac{y^2}{2} \right) dy \\
 & \Rightarrow \frac{1}{4} \left[\frac{y^2}{2} + y \right]_{-1/2}^1 - \frac{1}{6} \left(y^3 \right)_{-1/2}^1 \\
 & \Rightarrow \frac{1}{4} \left[\left(\frac{1}{2} + 1 \right) - \left(\frac{1}{8} - \frac{1}{2} \right) \right] - \frac{1}{6} \left[1 + \frac{1}{8} \right] \\
 & \Rightarrow \frac{1}{4} \left[\frac{3}{2} + \frac{3}{8} \right] - \frac{1}{6} \left[\frac{9}{8} \right] \quad \Rightarrow \quad \frac{15}{32} - \frac{6}{32} = \frac{9}{32}
 \end{aligned}$$

Q46. The area (in sq. units) of the region $\{(x, y) : y^2 \geq 2x \text{ and } x^2 + y^2 \geq 4x, x \geq 0, y \geq 0\}$ is :

- (A) $\pi - \frac{4\sqrt{2}}{3}$ (B) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$ (C) $\pi - \frac{4}{3}$ (D) $\pi - \frac{8}{3}$

Ans. D

Sol.



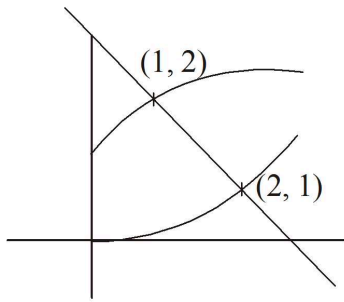
$$\begin{aligned}
 \text{Area} &= \frac{1}{4} (\text{Area of circle}) - \int_0^2 \sqrt{2x} dx \\
 &= \pi - \frac{8}{3}
 \end{aligned}$$

Q47. The area (in sq. units) of the region $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y \text{ and } y \leq 1 + \sqrt{x}\}$ is

- (A) $\frac{7}{3}$ (B) $\frac{5}{2}$ (C) $\frac{59}{12}$ (D) $\frac{3}{2}$

Ans. B

Sol.



$$\begin{aligned} \text{Area} &= \int_0^1 \left[\left(1 + \sqrt{x}\right) - \frac{x^2}{4} \right] dx + \int_1^2 \left[\left(3 - x\right) - \frac{x^2}{4} \right] dx \\ &= \frac{5}{2} \end{aligned}$$

Q48. If A is the area between the curve $y = \sin x$ and x-axis in the interval $[0, \pi/2]$, then the area between $y = \sin 2x$ and x-axis in this interval will be-

- (A) A (B) 2A (C) A/2 (D) None of these

Ans. A

Sol. -

Q49. If $(a, 0)$; $a > 0$ is the point where the curve $y = \sin 2x - \sqrt{3} \sin x$ cuts the x-axis first, A is the area bounded by this part of the curve, the origin and the positive x-axis, then

- (A) $4A + 8 \cos a = 7$ (B) $4A + 8 \sin a = 7$ (C) $4A - 8 \sin a = 7$ (D) $4A - 8 \cos a = 7$

Ans. A

Sol. $(a, 0)$ lies on the given curve

$$\therefore 0 = \sin 2a - \sqrt{3} \sin a \Rightarrow \sin a = 0 \text{ or } \cos a = \sqrt{3}/2$$

$$\Rightarrow a = \frac{\pi}{6} \text{ (as } a > 0 \text{ and the first point of intersection with positive X-axis)}$$

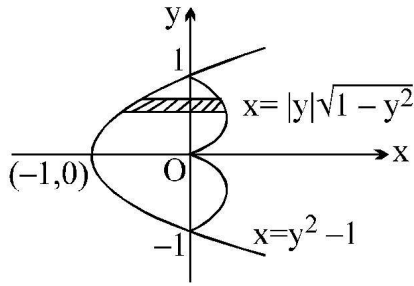
$$\begin{aligned} \text{and } A &= \int_0^{\pi/6} (\sin 2x - \sqrt{3} \sin x) dx = \left(-\frac{\cos 2x}{2} + \sqrt{3} \cos x \right)_0^{\pi/6} \\ &= \left(-\frac{1}{4} + \frac{3}{2} \right) - \left(-\frac{1}{2} + \sqrt{3} \right) = \frac{7}{4} - \sqrt{3} = \frac{7}{4} - \cos a \Rightarrow 4A + 8 \cos a = 7 \end{aligned}$$

Q50. Area of the region enclosed between the curves $x = y^2 - 1$ and $x = |y| \sqrt{1 - y^2}$ is

- (A) 1 (B) 4/3 (C) 2/3 (D) 2

Ans. D

Sol.



$$A = 2 \int_{-1}^0 \left[y\sqrt{1-y^2} - (y^2 - 1) \right] dy$$

$$= 2$$

Q51. The area bounded by the curve $y = f(x)$, the x -axis & the ordinates $x = 1$ & $x = b$ is $(b - 1) \sin(3b + 4)$. Then $f(x)$ is:

- (A) $(x - 1) \cos(3x + 4)$ (B) $\sin(3x + 4)$ (C) $\sin(3x + 4) + 3(x - 1) \cdot \cos(3x + 4)$
 (D) none

Ans. C

Sol. -

Q52. If the area bounded by the x -axis, curve $y = f(x)$ and the lines $x = 1$, $x = b$ is equal to $\sqrt{b^2 + 1} - \sqrt{2}$ for all $b > 1$, then $f(x)$ is

- (A) $\sqrt{x - 1}$ (B) $\sqrt{x + 1}$ (C) $\sqrt{x^2 + 1}$ (D) $\frac{x}{\sqrt{1 + x^2}}$

Ans. D

Sol. -

Q53. Let $f(x)$ be a non-negative continuous function such that the area bounded by the curve $y = f(x)$, x -axis and the ordinates $x = \frac{\pi}{4}$ and $x = \beta > \frac{\pi}{4}$ is

$\left(\beta \sin \beta + \frac{\pi}{4} \cos \beta + \sqrt{2} \beta \right)$. Then $f\left(\frac{\pi}{2}\right)$ is -

- (A) $\left(\frac{\pi}{4} + \sqrt{2} - 1 \right)$ (B) $\left(\frac{\pi}{4} - \sqrt{2} + 1 \right)$ (C) $\left(1 - \frac{\pi}{4} - \sqrt{2} \right)$ (D) $\left(1 - \frac{\pi}{4} + \sqrt{2} \right)$

Ans. D

Sol. -

Exercise J02

Q1. Find the area bounded on the right by the line $x + y = 2$, on the left by the parabola $y = x^2$ and below by the x-axis in first quadrant.

Ans. $5/6$ sq. units

Sol. -

Q2. Find the area of the region bounded by curves $f(x) = (x - 4)^2$, $g(x) = 16 - x^2$ and the x-axis.

Ans. 64

Sol. -

Q3. Compute the area of the region bounded by the curves $y = e \cdot x \cdot \ln x$ & $y = \ln x / (e \cdot x)$ where $\ln e = 1$.

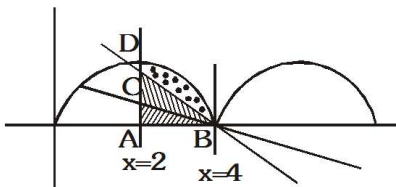
Ans. $(e^2 - 5)/4$ e sq. units

Sol. -

Q4. A figure is bounded by the curves $y = |\sqrt{2} \sin \frac{\pi x}{4}|$, $y = 0$, $x = 2$ & $x = 4$. At what angles to the positive x-axis straight lines must be drawn through $(4, 0)$ so that these lines partition the figure into three parts of the same size.

Ans. $\pi - \tan^{-1} \frac{2\sqrt{2}}{3\pi}$; $\pi - \tan^{-1} \frac{4\sqrt{2}}{3\pi}$

Sol. Let equation of line is $y = mx - 4m$



$$A = \int_2^4 \sqrt{2} \sin \frac{\pi x}{4} dx = \left[-\sqrt{2} \frac{4}{\pi} \cos \frac{\pi x}{4} \right]_2^4 = \frac{4\sqrt{2}}{\pi} \quad \dots\dots\dots(i)$$

$$\text{Also area of } \triangle ABC = 2 \cdot 2 \cdot (-2m_1) = -2m_1 \quad \dots\dots\dots(ii)$$

from (i) and (ii)

$$-2m_1 = \frac{4\sqrt{2}}{3\pi} \Rightarrow m_1 = \frac{-2\sqrt{2}}{3\pi}$$

$$\Rightarrow \tan(\pi - \theta_1) = \frac{-2\sqrt{2}}{3\pi} \Rightarrow \pi - \theta_1 = \tan^{-1} \frac{2\sqrt{2}}{3\pi}$$

$$\Rightarrow \theta_1 = \pi - \tan^{-1} \frac{2\sqrt{2}}{3\pi} \text{ or } \frac{1}{2} \cdot (2) \cdot (-2m_2) = \frac{8\sqrt{2}}{3\pi}$$

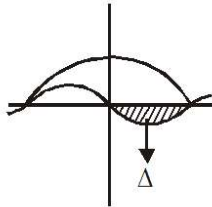
$$\Rightarrow m_2 = \frac{-4\sqrt{2}}{3\pi} \Rightarrow \tan(\pi - \theta_2) = \frac{-4\sqrt{2}}{3\pi}$$

$$\Rightarrow \theta_2 = \pi \tan^{-1} \frac{4\sqrt{2}}{3\pi}$$

Q5. Find the area bounded by the curves $y = \sqrt{1-x^2}$ and $y = x^3 - x$. Also find the ratio in which the y-axis divided this area.

Ans. $\frac{\pi}{2}; \frac{\pi-1}{\pi+1}$

Sol.



$$\Delta \left| \int_0^1 (x^3 - x) dx \right| = \frac{1}{4}$$

$$\text{Area} = \frac{\pi}{4} + \frac{1}{4} + \frac{\pi}{4} - \frac{1}{4} = \frac{\pi}{2}$$

$$\text{required ratio} = \frac{\frac{\pi}{4} + \frac{1}{4}}{\frac{\pi}{4} - \frac{1}{4}}$$

Q6. The line $3x + 2y = 13$ divides the area enclosed by the curve, $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ into two parts. Find the ratio of the larger area to the smaller area.

Ans. $\frac{3\pi+1}{\pi-2}$

Sol. -

Q7. Find the area enclosed between the curves : $y = \log_e(x+e)$, $x = \log_e(1/y)$ & the x-axis.

Ans. 2 sq. units

Sol. -

Q8. Compute the area of the curvilinear triangle bounded by the y-axis & the curve, $y = \tan x$ & $y = (2/3)\cos x$.

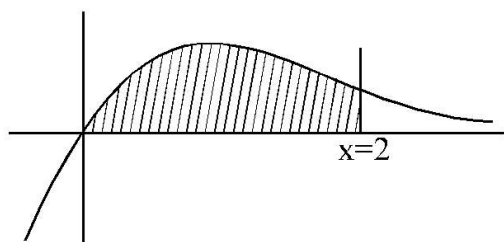
Ans. $\frac{1}{3} + \ln\left(\frac{\sqrt{3}}{2}\right)$ sq. units

Sol. -

Q9. Find the area bounded by the curve $y = x e^{-x}$; $xy = 0$ and $x = c$ where c is the x-coordinate of the curve's inflection point.

Ans. $1 - 3e^{-2}$

Sol.



$$y = xe^{-x}$$

$$y' = e^{-x} - x e^{-x} = (1 - x)e^{-x} \uparrow \text{ for } x < 1$$

$$y'' = -e^{-x} - [e^{-x} - x e^{-x}] = e^{-x} [-1 - 1 + x]$$

$$= (x - 2)e^{-x}$$

$$\text{for point inflection } y'' = 0 \Rightarrow x = 2$$

$$A = \int_0^2 x e^{-x} dx = -x e^{-x} \Big|_0^2 + \int_0^2 e^{-x}$$

$$= (-2 e^{-2}) - \left(e^{-x} \right)_0^2$$

$$= -e^{-2} - (e^{-2} - 1) = 1 - e^{-2} - 2e^{-2} = 1 - 3e^{-2} \text{ Ans}$$

Q10. Find the area bounded by the curve $y = e^{-x^2}$, the x-axis, and the line $x = c$ where $y(c)$ is maximum.

Ans. $\frac{1}{2}(1 - e^{-1/2})$

Sol. -

Q11. Consider a circle $x^2 + (y - 1)^2 = 1$ and the parabola $y = -\frac{x^2}{4}$. The common tangents to the two curves constitute a triangle ABC, the point A and B being on the x-axis and C on the y-axis. CA produced touches the parabola at P and CB produced touches the parabola at Q.

(a) Find the equation of the common tangent BC.

(b) Find the area of the portion between the upper arc of the circle and the common tangents QC and PC.

(c) Find the area enclosed by the parabola $y = -\frac{x^2}{4}$, the x-axis and the lines AP and BQ.

Ans. (a) $\sqrt{3}x - y + 3 = 0$; (b) $\left(\sqrt{3} - \frac{\pi}{3} \right)$; (c) $\sqrt{3}$

Sol. -

- Q12. If $f(x)$ is monotonic in (a, b) then prove that the area bounded by the ordinates at $x = a$; $x = b$; $y = f(x)$ and $y = f(c)$, $c \in (a, b)$ is minimum when $c = \frac{a+b}{2}$.

Hence if the area bounded by the graph of $f(x) = \frac{x^3}{3} - x^2 + a$, the straight lines $x = 0$, $x = 2$ and the x-axis is minimum then find the value of 'a'.

Ans. $a = \frac{2}{3}$

Sol. -

- Q13. Find the area of the region bounded by the curves $y = x^2$, $y = |2 - x^2|$ and $y = 2$, which lies to the right of the line $x = 1$.

Ans. $\left(\frac{30}{3} - 4\sqrt{2}\right)$ sq. units

- Sol. The given curves are $y = x^2$ which is an upward parabola with vertex at $(0, 0)$

$$y = |2 - x^2|$$

$$\text{or } y = \begin{cases} 2 - x^2 & \text{if } -\sqrt{2} < x < \sqrt{2} \\ x^2 - 2 & \text{if } x < -\sqrt{2} \text{ or } x > \sqrt{2} \end{cases}$$

$$\text{or } x^2 = -(y - 2); \quad -\sqrt{2} < x < \sqrt{2} \quad \dots\dots\dots(2)$$

a downward parabola with vertex at $(0, 2)$

$$x^2 = y + 2; \quad x < -\sqrt{2}, x > \sqrt{2} \quad \dots\dots\dots(3)$$

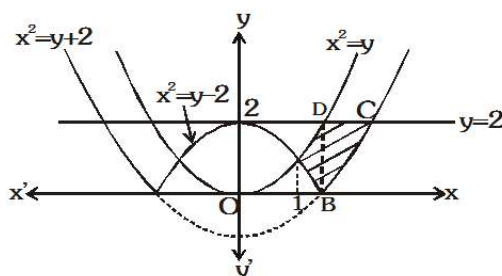
On upward parabola with vertex at $(0, -2)$

$$y = 2 \quad \dots\dots\dots(4)$$

Straight line parallel to x-axis

$$x = 1 \quad \dots\dots\dots(5)$$

The graph of these curves is as follows.



$\therefore$ Required area = BCDEB

$$= \int_1^{\sqrt{2}} [x^2 - (2 - x^2)] dx + \int_{\sqrt{2}}^2 [2x - (x^2 - 2)] dx$$

$$= \int_1^{\sqrt{2}} (2x^2 - 2) dx + \int_{\sqrt{2}}^2 (4 - x^2) dx = \left(\frac{20}{3} 4\sqrt{2} \right) \text{ sq. units}$$

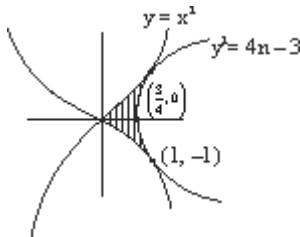
Q14. Find the area bounded by the curves $x^2 = y$, $x^2 = -y$ and $y^2 = 4x - 3$.

Ans. $\frac{1}{3} \text{ sq. units}$

Sol. $x^2 = y$
 $x^2 = -y$
 $y^2 = x$
 POI
 $y^2 = 4x - 3$ and $x^2 = y$
 $x^4 = 4x - 3$
 $x^4 - 4x + 3 = 0$
 $(x - 1)(x^3 + x^2 + x - 3) =$

$$(x - 1)(x - 1) \left(\underbrace{x^2 + 2x + 3}_{D < 0} \right) = 0$$

$n = 1$



$$\begin{aligned} A &= 2 \left[\int_0^1 x^2 dx - \int_{\frac{3}{4}}^1 \sqrt{4x - 3} dx \right] \\ &= 2 \left[\frac{x^3}{3} \Big|_0^1 - \frac{2}{3} \frac{(4x - 3)^{\frac{3}{2}}}{4} \Big|_{\frac{3}{4}}^1 \right] \\ &= 2 \left(\frac{1}{3} - \frac{1}{6} \right) \\ &= 2 \times \frac{1}{6} = \frac{1}{3} \end{aligned}$$

Q15. Let $f(x)$ be a quadratic polynomial and a, b, c be distinct real numbers such that

$$\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2 & + & 3a \\ 3b^2 & + & 3b \\ 3c^2 & + & 3c \end{bmatrix}.$$

Let V be the point of maximum of the curve $y = f(x)$. If A and B are the points on this curve such that the curve meets the positive x-axis at A and the chord AB subtends a right angle at V, then find the area enclosed by the curve and the chord AB.

Ans. (a) $\frac{1}{3}$ sq. units ; (b) $\frac{125}{3}$ sq. units

Sol. We have,
$$\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2 + 3a \\ 3b^2 + 3b \\ 3c^2 + 3c \end{bmatrix}$$

$$\Rightarrow 4a^2 |(-1) + 4a| (1) + | (2) = 3a^2 + 3a$$

$$4b^2 |(-1) + 4b| (1) + | (2) = 3b^2 + 3b$$

$$4c^2 |(-1) + 4c| (1) + | (2) = 3c^2 + 3c$$

Consider the equation

$$4x^2 |(-1) + 4x| (1) + | (2) = 3x^2 + 3x$$

$$\text{or } [4 |(-1) - 3]x^2 + [4| (1) - 3]x + |(2) = 0$$

Then clearly this equation is satisfied by

$$x = a, b, c$$

A quadratic equation satisfied by more than two values of x means it is an identity and hence

$$4|(-1) - 3 = \Rightarrow |(-1) = 3/4$$

$$4| (1) - 3 = 0 \Rightarrow | (1) = 3/4$$

$$| (2) = 0 \Rightarrow | (2) = 0$$

Let $| (x) = px^2 + qx + r$ [$| (x)$ being a quad. equation]

$$|(-1) = \frac{3}{4} \Rightarrow p - q + r = \frac{3}{4}$$

$$| (1) = \frac{3}{4} \Rightarrow p + q + r = \frac{3}{4}$$

$$| (2) = 0 \Rightarrow 4p + 2q + r = 0$$

Solving the above we get $q = 0, p = \frac{-1}{4}, r = 1$

$$\therefore | (x) = -\frac{1}{4} x^2 + 1$$

It's maximum value occur at $|' (x) = 0$

$$\text{i.e., } x = 0 \text{ then } | (x) = 1 \quad \therefore V(0, 1)$$

A $(-2, 0)$ is the pt. where curve meet x-axis

Let B be the pt. $\left(h, \frac{4 - h^2}{4}\right)$

As $\angle AVB = 90^\circ$

$$m_{AV} \cdot m_{BV} = -1$$

$$\Rightarrow \frac{1}{2} \times \left(\frac{-h}{4} \right) = -1$$

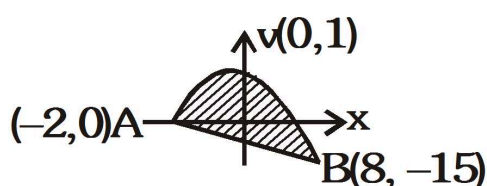
$$\Rightarrow h = 8 \quad \therefore B(8, -15)$$

Equation of chord AB is

$$y + 15 = \frac{0 - (-15)}{-2 - 8} (x - 8)$$

$$\Rightarrow 3x + 2y + 6 = 0$$

Required area is the area of shaded region given by



$$= \int_{-2}^8 \left[\left(-\frac{x^2}{4} + 1 \right) - \left(\frac{-6 - 3x}{2} \right) \right] dx$$

$$= \frac{125}{3} \text{ sq. units}$$

Q16. If the area enclosed by the parabolas $y = a - x^2$ and $y = x^2$ is $18\sqrt{2}$ sq. units. Find the value of 'a'.

Ans. $a = 9$

Sol. -

Q17. Find the values of m ($m > 0$) for which the area bounded by the line $y = mx + 2$ and $x = 2y - y^2$ is

(i) $9/2$ square units & (ii) minimum. Also find the minimum area.

Ans. (i) $m = 1$, (ii) $m = \infty$; $A_{\min} = 4/3$

Sol. -

Q18. Consider two curves $C_1 : y = \frac{1}{2}$ and $C_2 : y = \ln x$ on the xy plane. Let D_1 denotes the region surrounded by

C_1 , C_2 and the line $x = 1$ and D_2 denotes the region surrounded by C_1 , C_2 and the line $x = a$. If $D_1 = D_2$. Find the value of 'a'.

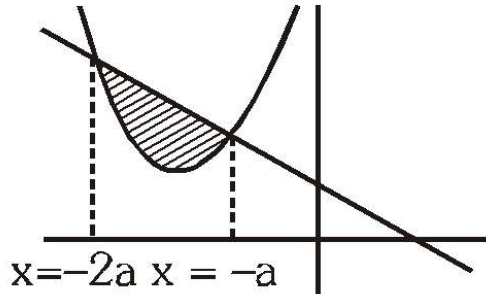
Ans. e

Sol. -

Q19. Find the value (s) of the parameter 'a' ($a > 0$) for each of which the area of the figure bounded by the straight line, $y = \frac{a^2 - ax}{1 + a^4}$ & the parabola $y = \frac{x^2 + 2ax + 3a^2}{1 + a^4}$ is the greatest.

Ans. $a = 3^{1/4}$

Sol.



$$A = \int_{-a}^{-2a} \frac{a^2 - ax(x^2 + 2ax + 3a^2)}{1 + a^4} dx$$

$$= \frac{3}{2} \frac{a^3}{1 + a^4}$$

$$\text{Now } f(a) = \frac{3}{2} \frac{a^3}{1 + a^4}$$

$$\Rightarrow f'(a) = 0$$

$$\Rightarrow (1 + a^4) 3a^2 - a^3 4a^3 = 0$$

$$\Rightarrow a_{\min} = 0, a_{\max} = 3^{1/4}$$

Q20. For what value of 'a' is the area bounded by the curve $y = a^2x^2 + ax + 1$ and the straight line $y = 0$, $x = 0$ & $x = 1$ the least ?

Ans. $a = -3/4$

Sol. -

Q21. Find the positive value of 'a' for which the parabola $y = x^2 + 1$ bisects the area of the rectangle with vertices $(0, 0)$, $(a, 0)$, $(0, a^2 + 1)$ and $(a, a^2 + 1)$.

Ans. $\sqrt{3}$

Sol. -

Q22. Find the value of 'c' for which the area of the figure bounded by the curve, $y = 8x^2 - x^5$, the straight lines $x = 1$ & $x = c$ & the abscissa axis is equal to $16/3$.

Ans. $C-1 \text{ or } (8 - \sqrt{17})^{1/3}$

Sol. -

Q23. Consider one side AB of a square ABCD, (read in order) on the line $y = 2x - 17$, and the other two vertices

C, D on the parabola $y = x^2$.

(a) Find the minimum intercept of the line CD on y-axis.

(b) Find the maximum possible area of the square ABCD.

(c) Find the area enclosed by the line CD with minimum y-intercept and the parabola $y = x^2$.

Ans. (a) 3; (b) 1280; (c) $\frac{32}{3}$

Sol. -

Paragraph

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of $f(x)$ and let $t = |s|$.

\n

Q24. The real number s lies in the interval

(A) $\left(-\frac{1}{4}, 0\right)$ (B) $\left(-11, -\frac{3}{4}\right)$ (C) $\left(-\frac{3}{4} - \frac{1}{2}\right)$ (D) $\left(0, \frac{1}{4}\right)$

Ans. C

Sol. $\therefore f(x) = 2 + 6x + 12x^2 > 0 \forall x \in \mathbb{R}$

$\therefore f(x)$ is strictly increasing in $\mathbb{R}$

$\therefore f(0) = 1, f(-1) = -2, f\left(-\frac{1}{2}\right) = \frac{1}{4} \text{ \& } f\left(-\frac{3}{4}\right) = -\frac{1}{2}$

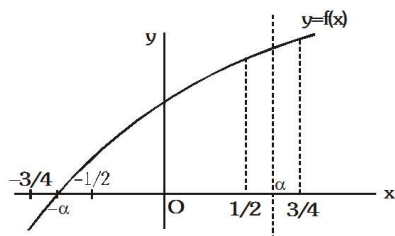
$\therefore f(x) = 0$ has only one real root lying in $\left(-\frac{3}{4}, -\frac{1}{2}\right)$

Q25. The area bounded by the curve $y = f(x)$ and the lines $x = 0, y = 0$ and $x = t$, lies in the interval

(A) $\left(\frac{3}{4}, 3\right)$ (B) $\left(\frac{21}{64}, \frac{11}{16}\right)$ (C) (9, 10) (D) $\left(0, \frac{21}{64}\right)$

Ans. A

Sol.



Let real root is $-\alpha$

$\Rightarrow t = |s| = \alpha$

Required area

$$A = \int_0^{\alpha} f(x) dx \quad \& \quad \int_0^{1/2} f(x) dx < A < \int_0^{3/4} f(x) dx$$

$$\Rightarrow \left| x + x^2 + x^3 + x^4 \right|_0^{1/2} < A < \left| x + x^2 + x^3 + x^4 \right|_0^{3/4} < \left| 4x \right|_0^{3/4}$$

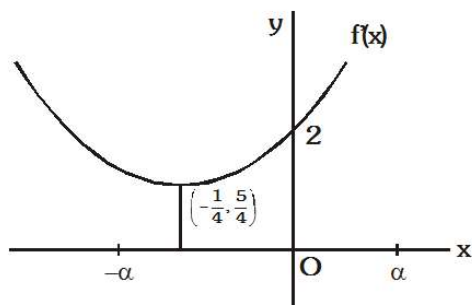
$$\Rightarrow \frac{15}{16} < A < 3$$

Q26. The function $f'(x)$ is

- (A) increasing in $\left(-t, -\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4}, t\right)$
 (B) decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$ (C) increasing in $(-t, t)$
 (D) decreasing in $(-t, t)$

Ans. B

Sol.



$$f'(x) = 2(6x^2 + 3x + 1)$$

$$\Rightarrow f'(x) \text{ is decreasing in } \left(-\alpha, -\frac{1}{4}\right) \text{ increasing in } \left(-\frac{1}{4}, \alpha\right)$$

$$\text{or } f'(x) \text{ is decreasing in } \left(-t, -\frac{1}{4}\right) \text{ increasing in } \left(-\frac{1}{4}, t\right)$$

Paragraph

Consider three real-valued functions f , g and h defined on $\mathbb{R}$ (the set of real numbers).

Let $f(x) = 2x^3 + 3\left(1 - \frac{3a}{2}\right)x^2 + 3(a^2 - a)x + b$ where $a, b \in \mathbb{R}$ and $g(x) = \frac{f'(x)}{6}$.

Also $h(x)$ is such that $h''(x) = 6x - 4$ and $h(x)$ has a local minimum value 5 at $x = 1$.

Q27. The true set of values of a for which $f(x)$ has negative point of local minimum, is

- (A) $(-\infty, 0)$ (B) $(1, \infty)$ (C) $(0, 1)$ (D) $(1, \infty) - \{2\}$

Ans. A

Sol. We have

$$f(x) = 2x^3 + 3 \left(1 - \frac{3a}{2}\right) x^2 + 3(a^2 + a)x + b$$

$$\text{so, } f'(x) = 6x^2 + 6 \left(1 - \frac{3a}{2}\right) x + 3(a^2 - a)$$

Now, for $f(x)$ to have negative point of local minimum, we must have both roots of the equation $f'(x) = 0$ real (unequal) and negative.

$$\therefore \text{Discriminant} > 0 \Rightarrow 36 \left[\left(1 - \frac{3a}{2}\right)^2 - 2(a^2 - a) \right] > 0 \Rightarrow \left(\frac{a}{2} - 1\right)^2 > 0 \Rightarrow a \in \mathbb{R} - \{2\}$$

.....(1)

$$\text{Also, sum of roots} < 0 \Rightarrow \frac{-6 \left(1 - \frac{3a}{2}\right)}{6} < 0 \Rightarrow \frac{3a}{2} - 1 < 0 \Rightarrow a < \frac{2}{3}$$

.....(2)

$$\text{And product of roots} > 0 \Rightarrow \frac{3(a^2 - a)}{6} > 0 \Rightarrow a(a - 1) > 0 \Rightarrow a \in (-\infty, 0) \cap (1, \infty)$$

.....(3)

$\therefore$ From (1) $\cap$ (2) $\cap$ (3), we get $a \in (-\infty, 0)$

Q28. The complete set of values of a for which vertex of parabola $y = g(x)$ has negative ordinate, is

(A) $\left(\frac{1}{2}, 1\right)$ (B) $(0, \infty)$ (C) $\mathbb{R} - \{2\}$ (D) $(1, \infty) - \{2\}$

Ans. C

Sol. We have

$$g(x) = \frac{f'(x)}{6} = x^2 + \left(1 - \frac{3a}{2}\right)x + \left(\frac{a^2 - a}{2}\right)$$

$$\text{As discriminant of } g(x) = \left(\frac{a}{2} - 1\right)^2 > 0 \Rightarrow a \in \mathbb{R} - \{2\}$$

$\Rightarrow$ Ordinate of the vertex of parabola $y = g(x)$ lies below x-axis.

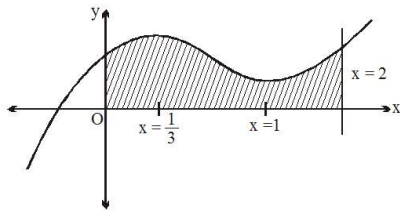
So, $a \in \mathbb{R} - \{2\}$

Q29. The area bounded by $y = h(x)$ between $x = 0$ and $x = 2$, is

(A) $\frac{23}{3}$ (B) $\frac{20}{3}$ (C) $\frac{40}{3}$ (D) $\frac{32}{3}$

Ans. D

Sol.



We have

$$h''(x) = 6x - 4 \Rightarrow h'(x) = 3x^2 - 4x + c$$

$$\text{As } h'(1) = 0, \text{ so } 0 = 3 - 4 + c \Rightarrow c = 1$$

$$\therefore h'(x) = 3x^2 - 4x + 1 \Rightarrow h(x) = x^3 - 2x^2 + x + k$$

$$\text{As } h'(2) = 5, \text{ so } 5 = 12 - 8 + 1 + k \Rightarrow k = 5$$

$$\text{Hence, } h(x) = x^3 - 2x^2 - x + 5$$

Clearly, required area

$$= \int_{1/3}^2 h(x) dx = \int_{1/3}^2 (x^3 - 2x^2 - x + 5) dx = \left(\frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} + 5x \right)_{1/3}^2 = \frac{32}{3} \text{ (square units)}$$

Paragraph

Consider the functions defined implicitly by the equation $y^3 - 3y + x = 0$ on various intervals in the real line. If

$\forall x \in (-\infty, -2) \cup (2, \infty)$, the equation implicitly defines a unique real valued differentiable function $y = f(x)$. If $x \in (-2, 2)$, the equation implicitly defines a unique real valued differentiable function $y = g(x)$ satisfying $g(0) = 0$.

\n

Q30. If $f(-10\sqrt{2}) = 2\sqrt{2}$, then $f''(-10\sqrt{2}) =$

(A) $\frac{4\sqrt{2}}{7^3 3^2}$ (B) $-\frac{4\sqrt{2}}{7^3 3^2}$ (C) $\frac{4\sqrt{2}}{7^3 3}$ (D) $-\frac{4\sqrt{2}}{7^3 3}$

Ans. B

Sol. $y^3 - 3y + x = 0$

$$3y^2 y' - 3y' + 1 = 0 \quad y' = \frac{-1}{3(y^2 - 1)}$$

$$f(-10\sqrt{2}) = 2\sqrt{2}$$

$$f'(-10\sqrt{2}) = \frac{1}{3(7)} = -\frac{1}{21}$$

$$6y(y')^2 + 3y^2 y'' - 3y'' = 0$$

$$y'' = -\frac{2y(y')^2}{y^2 - 1}$$

$$f'(-10\sqrt{2}) = \frac{-2(2\sqrt{2})}{441 \times 1} = \frac{-4\sqrt{2}}{7^3 3^2}$$

Q31. The area of the region bounded by the curve $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$, where $-\infty < a < b < -2$, is

$$\begin{array}{ll} \text{(A)} \int_a^b \frac{x}{3(f(x))^2 - 1} dx + b f(b) - a f(a) & \text{(B)} - \int_a^b \frac{x}{3(f(x))^2 - 1} dx + b f(b) - a f(a) \\ \text{(C)} \int_a^b \frac{x}{3(f(x))^2 - 1} dx - b f(b) + a f(a) & \text{(D)} - \int_a^b \frac{x}{3(f(x))^2 - 1} dx - b f(b) + a f(a) \end{array}$$

Ans. A

$$\begin{aligned} \text{Sol. } \int_a^b f(x) dx &= \left[x f(x) \right]_a^b - \int_a^b (x) f'(x) dx \\ &= b f(b) - a f(a) + \int_a^b \frac{x}{3[(f(x))^2 - 1]} dx \\ &= \int_a^b \frac{x}{3[(f(x))^2 - 1]} dx + b f(b) - a f(a) \end{aligned}$$

Q32. $\int_{-1}^1 g'(x) dx =$

(A) $2g(-1)$ (B) 0 (C) $-2g(1)$ (D) $2g(1)$

Ans. D

$$\text{Sol. } \int_{-1}^1 g'(x) dx = g(1) - g(-1)$$

$$\text{Now } g(1) = -g(-1)$$

(as $g'(x)$ is an even function)

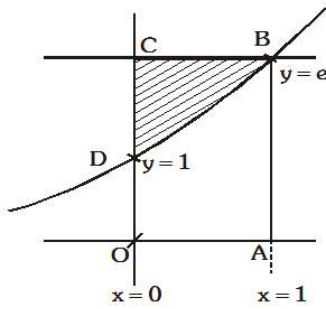
$$\text{so } \int_{-1}^1 g'(x) dx = 2g(1) = -2g(-1)$$

Q33. Area of the region bounded by the curve $y = e^x$ and lines $x = 0$ and $y = e$ is

$$\begin{array}{llll} \text{(A)} e - 1 & \text{(B)} \int_1^e \ln(e + 1 - y) dy & \text{(C)} e - \int_0^1 e^x dx & \text{(D)} \int_1^e \ln y dy \end{array}$$

Ans. D, C, B

Sol.



$$A = \int_1^e \ln y \, dy$$

Apply

$$= \int_1^e \ln (e + 1 - y) dy$$

$$A = \text{ar (OABC)} - \text{ar (OABD)}$$

$$= e - \int_1^e e^x dx$$

Q34. The area bounded by a curve, the axis of co-ordinates & the ordinate of some point of the curve is equal to the length of the corresponding arc of the curve . If the curve passes through the point P (0 , 1) then the equation of this curve can be :

(A) $y = \frac{1}{2}(e^x - e^{-x} + 2)$ (B) $y = \frac{1}{2}(e^x + e^{-x})$ (C) $y = 1$ (D) $y = \frac{2}{e^x + e^{-x}}$

Ans. C, B

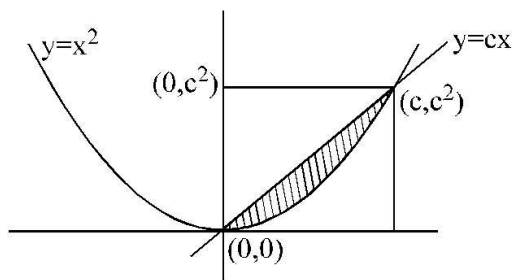
Sol. -

Q35. Let T be the triangle with vertices (0, 0), (0, c²) and (c, c²) and let R be the region between y = cx and y = x² where c > 0 then

(A) $\text{Area (R)} = \frac{c^3}{6}$ (B) $\text{Area (R)} = \frac{c^3}{3}$ (C) $\lim_{c \rightarrow 0^+} \frac{\text{Area(T)}}{\text{Area(R)}} = 3$ (D) $\lim_{c \rightarrow 0^+} \frac{\text{Area(T)}}{\text{Area(R)}} = \frac{3}{2}$

Ans. A, C

Sol.



$$\text{Area (T)} = \frac{c \cdot c^3}{2} = \frac{c^3}{2}$$

$$\text{Area (R)} = \frac{c^3}{2} - \int_0^c x^2 dx$$

$$= \frac{c^3}{2} - \frac{c^3}{3} = \frac{c^3}{6}$$

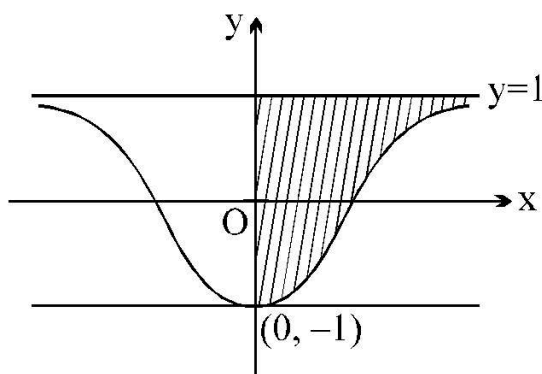
$$\therefore \lim_{x \rightarrow \infty^+} \frac{c^3}{2} \cdot \frac{6}{c^3} = 3$$

Q36. Suppose f is defined from $\mathbb{R} \rightarrow [-1, 1]$ as $f(x) = \frac{x^2 - 1}{x^2 + 1}$ where $\mathbb{R}$ is the set of real number. Then the statement which does not hold is

- (A) f is many one onto (B) f increases for $x > 0$ and decrease for $x < 0$
 (C) minimum value is not attained even though f is bounded
 (D) the area included by the curve $y = f(x)$ and the line $y = 1$ is π sq. units.

Ans. A, C, D

Sol.



$$y = f(x) = \frac{x^2 - 1}{x^2 + 1} = 1 - \frac{2}{x^2 + 1}$$

$$f'(x) = \frac{4x}{(x^2 + 1)^2}$$

$x > 0$, f is increasing and $x < 0$ f is decreasing $\Rightarrow$ (B) is true

range is $[-1, 1] \Rightarrow$ into $\Rightarrow$ (A) is false;

minimum value occurs at $x = 0$ and $f(0) = -1 \Rightarrow$ (C) is false

$$A = 2 \int_0^\infty \left(-\frac{x^2 - 1}{x^2 + 1} \right) dx \int_0^\infty \frac{dx}{x^2 + 1} = 4 \cdot \tan^{-1} \Big|_0^\infty = 4 \cdot \frac{\pi}{2} = 2\pi \Rightarrow \text{(D) is false}$$

Q37. Match the column type

Column I		Column II
(A) The area between the curves $y = 2x^4 - x^2$, the x-axis and	(P) $2 \sin 1$	

	vertical lines passing through its two minima is		
(B)	The area bounded by the curve $x = at^2$, $y = 2at$, and the x-axis ($1 \leq t \leq 3$) is	(Q)	$\frac{7}{120}$
(C)	The area of a circle centred at $(1, 2)$ and passing through $(4, 6)$ is	(R)	$\frac{104a^2}{3}$
(D)	The area bounded by $y = \cos x$, $y = 0$, $ x = 1$ is given by	(S)	25π

(A) A - Q, B - R, C - S, D - P (B) A - C, B - R, Q - S, D - P

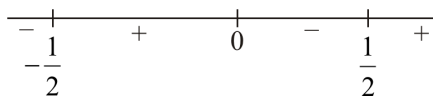
(C) A - Q, B - R, C - P, D - S (D) A - P, B - S, C - R, D - Q

Ans. A

Sol. (a) $y = 2x^4 - x^2$

$$y' = 8x^3 - 2x$$

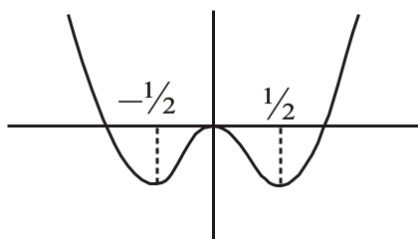
$$= 2x(4x^2 - 1)$$



$$\text{min at } x = -\frac{1}{2}, \frac{1}{2}$$

$$A = -2 \int_0^{\frac{1}{2}} (2x^4 - x^2) dx$$

$$= \frac{7}{120}$$



(b) Eliminating t , we obtain the equation of the curve as $y^2 = 4ax$

$$t = 1 \Rightarrow x = a \text{ and } t = 3 \Rightarrow x = 9a$$

Hence required area

$$= \int_a^{9a} y dx = \int_a^{9a} 2\sqrt{a}\sqrt{x} dx$$

$$= 2\sqrt{a} \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_a^{9a} = \frac{4}{3}\sqrt{2} \left[(9a)^{\frac{3}{2}} - a^{\frac{3}{2}} \right]$$

$$= \frac{4}{3}\sqrt{a} \left[27a^{\frac{3}{2}} - a^{\frac{3}{2}} \right] = \frac{4}{3}\sqrt{a} \left[26a^{\frac{3}{2}} \right] = \frac{104}{3}a^2$$

(c) Given that the centre of the circle is (1,2)

$$\text{Radius of the circle} = \sqrt{(4-1)^2 + (6-2)^2}$$

$$= \sqrt{9+16} = 5$$

So, the area of the circle = πr^2

$$= \pi \times (5)^2$$

$$= 25\pi$$

Q38.

Column I

Column II

(A) The area bounded by the curves $y = 2 \cos x$ and the x-axis from $x = 0$ to $x = 2\pi$ is

(P) $\frac{(\pi^3 - 8)}{4}$

(B) The area in the first quadrant between $x^2 + y^2 = \pi^2$ and $y = \sin x$ is

(Q) 1

(C) The measurement of the area bounded by the coordinate axes and the curve $y = \log_e x$ is

(R) $2a$

(D) The area of the loop between the curve $y = a \sin x$ and x-axis is

(S) 8

(A) A - P, B - S, C - Q, D - R (B) A - S, B - P, C - Q, D - R

(C) A - S, B - P, C - R, D - Q (D) A - R, B - Q, C - P, D - S

Ans. 5f38f8e75a4f9e3908e70324

Sol. -